

Halvor Hafnor

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EDUCATION

University of California - San Diego - San Diego, CA
Computer Engineering, M.S. (B.S. completed June 2026)

Major GPA: 3.894
Sep. 2024 – June 2027

Honors

- **Tau Beta Pi** — National engineering honor society.

De Anza College - Cupertino, CA
Computer Science & Engineering

GPA: 4.0
Sep. 2022 – June 2024

Academic and Technical Roles

- Math & Science Tutor and Computer Science Teaching Assistant, covering calculus, linear algebra, discrete math and physics, and programming in C++, Java and Assembly. Also a Technical Support Volunteer.

EXPERIENCE

Embedded ML Engineer
Aerendir Mobile

Jul. 2024 – Present
Mountain View, CA

- Developed embedded machine learning pipelines for classifying biometric motion signals from inertial sensors on **Linux**-based mobile devices.
- Implemented sensor interfaces and inference components using C++, Python, Bash, and QML.
- Built Android applications for real-time biometric data streaming and workflow control.
- Collected and classified the 400-participant motion dataset behind the company's biometric authentication work, roughly 40 hours of accelerometer and gyroscope data.
- Prototyped automotive driver monitoring from steering-wheel biometrics, shipped as **AerWheel** with the classifier running in the IMU's own firmware. Subject of a pending patent application naming me as an inventor.

Electrical Engineering Intern
ABB Electrification

Jun. 2021 – Sep. 2023
Skien, Norway

- Assembled and wired industrial electrical cabinets and control panels based on engineering schematics.
- Supported production and testing of custom electrical systems for industrial clients.
- Mentored incoming interns and contributed to improved onboarding efficiency.

Data Management Intern
Cathwell

Jun. 2019 – Aug. 2020
Langesund, Norway

- Supported migration of legacy data systems to updated platforms, ensuring data integrity and minimal downtime.

TECHNICAL SKILLS

Systems & Platforms: Embedded Linux, Android, QML/Qt

Languages: C++, C, Python, TypeScript, Java, Bash, ARM Assembly

ML & Vision: Machine Learning, Computer Vision, OpenCV, NumPy, SciPy

Tools: Git, GitHub, VS Code, Vim, SSH

PROJECTS

TabTerm — macOS terminal in Chrome tabs

- A background daemon owns the shell processes and Chrome owns only the views, so closing a tab kills nothing and reopening restores the process and the exact screen. **55,000 lines of TypeScript**, 514 tests.

Laser targeting on an autonomous car — UC San Diego, ECE 148

- **YOLOv8** with an **OAK-D** stereo camera on a Raspberry Pi 5, turning a detected person's pixel position and depth into pan and tilt angles so two servos hold a laser on them as they move.

Tryton — Underwater ROV

- Built the electrical subsystem of a modular underwater ROV: sensor wiring, LiPo power distribution and ESC integration, with a Pixhawk holding position from a depth sensor and IMU. Team of three.